ASSIGNMENT : 1

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Section: SB

Course: B.Tech

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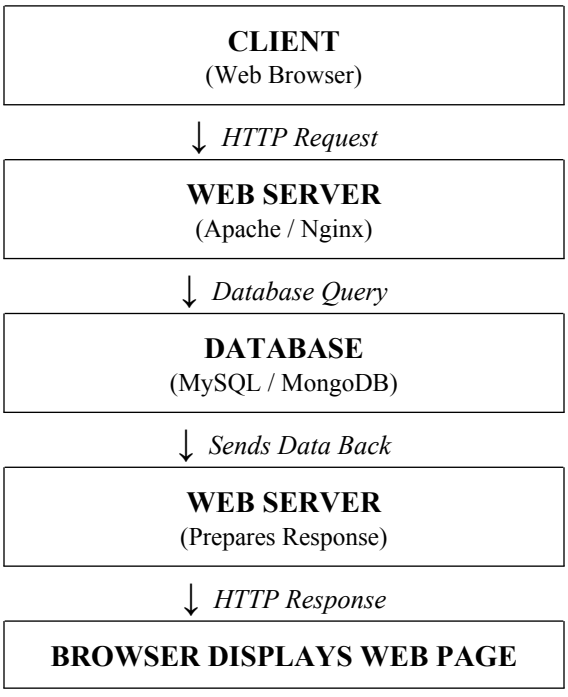
Submitted To: Mr. Akash Gupta

Q1. Explain the difference between frontend, backend, and full-stack development with suitable real-world examples.

Aspect	Frontend Development	Backend Development	Full-Stack Development
Focus	Works on the user interface (UI)	Works on the server, database, and application logic	Handles both frontend and backend development
Technologies	HTML, CSS, JavaScript	Python, Java, PHP, Node.js, etc.	Frontend and backend technologies together
Visibility	Users can directly see and interact with it	Users cannot directly see it	Can develop the complete application

Real-world example: An e-commerce website's product page layout and checkout form (frontend) work together with the order-processing, payment, and inventory logic running on the server (backend). A full-stack developer builds and connects both sides.

Q2. Create a simple diagram showing how the client-server model works in web architecture.



Q3. Describe how a browser requests and displays a web page from a web server.

1. User enters a website URL in the browser's address bar.
2. The browser sends an HTTP request to the web server.

3. The server receives the request.
4. The server processes the request.
5. If required, the server retrieves data from the database.
6. The server sends an HTTP response containing HTML, CSS, JavaScript, and images.
7. The browser downloads these files.
8. The browser renders the webpage and displays it to the user.

Q4. Identify and list the tools required to set up a web development environment. Explain the purpose of each.

Tool	Purpose
Visual Studio Code	Writing and editing code
Google Chrome	Testing web pages
Live Server Extension	Automatically refreshes web pages during development
Git	Version control
GitHub	Stores and shares code online
Node.js	Runs JavaScript outside the browser
npm	Installs JavaScript packages
Web Browser Developer Tools	Debugging HTML, CSS, and JavaScript

Q5. Explain what a web server is and give examples of commonly used servers.

A web server is software that receives requests from web browsers and sends back web pages or other resources over HTTP/HTTPS.

Common Web Servers

- Apache HTTP Server
- Nginx
- Microsoft IIS
- LiteSpeed
- Apache Tomcat (for Java applications)

Q6. Define the roles of a frontend developer, backend developer, and database administrator in a project.

Frontend Developer

- Designs the user interface.
- Creates responsive web pages.
- Uses HTML, CSS, JavaScript, React, Angular, etc.

Backend Developer

- Develops server-side logic.
- Handles authentication.
- Connects applications with databases.
- Uses Java, Python, PHP, Node.js, etc.

Database Administrator (DBA)

- Designs databases.
- Maintains database security.
- Performs backups and recovery.
- Optimizes database performance.

Q7. Install VS Code and configure it for HTML, CSS, and JavaScript development. Take a screenshot of the setup.

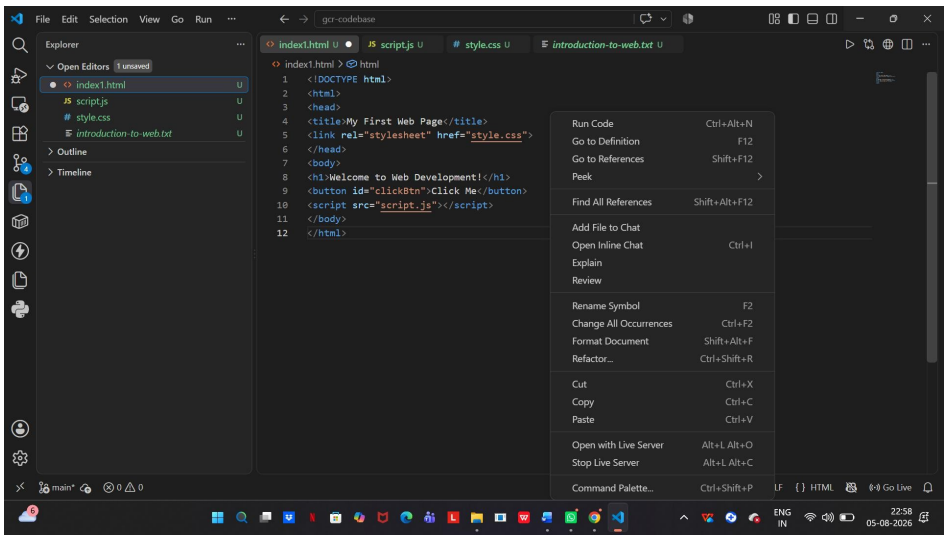


Figure 1 — VS Code setup for web development

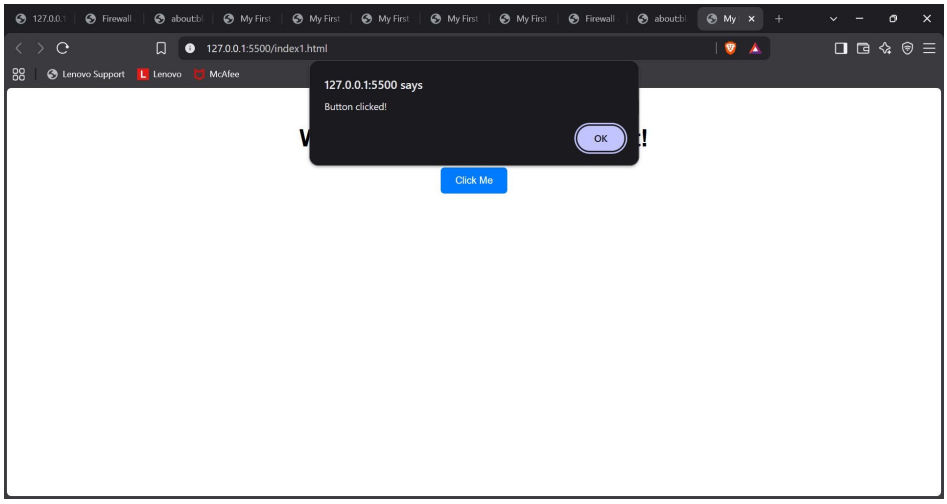


Figure 2 — VS Code configured with extensions

Q8. Explain the difference between static and dynamic websites. Provide an example of each.

Aspect	Static Website	Dynamic Website
Content	Displays the same content to every user; fixed and does not change automatically	Displays different content based on the user or data; changes automatically
Built With	HTML, CSS, and JavaScript	HTML, CSS, JavaScript, and backend languages (PHP, Python, Node.js) with databases
Development	Faster and easier to develop	More interactive but requires more development and maintenance
Suitability	Small websites such as portfolios or company information pages	Websites such as shopping sites, social media, and banking applications

Example: A static portfolio website built with plain HTML/CSS vs. a dynamic e-commerce site (e.g., Amazon) that personalizes content per user.

Q9. Research and list five web browsers. Explain how rendering engines differ between them.

Web Browser	Rendering Engine	Description
Google Chrome	Blink	Fast rendering engine developed by Google.
Mozilla Firefox	Gecko	Open-source rendering engine developed by Mozilla.
Microsoft Edge	Blink	Uses the same Blink engine as Chrome, making it compatible with most websites.
Apple Safari	WebKit	Rendering engine developed by Apple, mainly used on macOS and iOS.
Opera	Blink	Uses Blink and includes additional built-in features like VPN and ad blocker.

How Rendering Engines Differ

A rendering engine is the part of a web browser that reads HTML, CSS, and JavaScript and displays a web page on the screen.

- Blink (Chrome, Edge, Opera)
  - Fast and widely used.
  - Supports the latest web technologies.
  - Offers excellent compatibility with modern websites.
- Gecko (Firefox)

- Open-source and focused on web standards.
- Strong privacy and security features.
- May render some websites slightly differently from Blink.

- WebKit (Safari)
  - Optimized for Apple devices.
  - Provides good performance and energy efficiency on macOS and iOS.
  - Used mainly in Safari and some iOS browsers.

Q10. Draw a labeled diagram showing the basic web architecture flow — client, server, database, and APIs.

Basic Web Architecture

